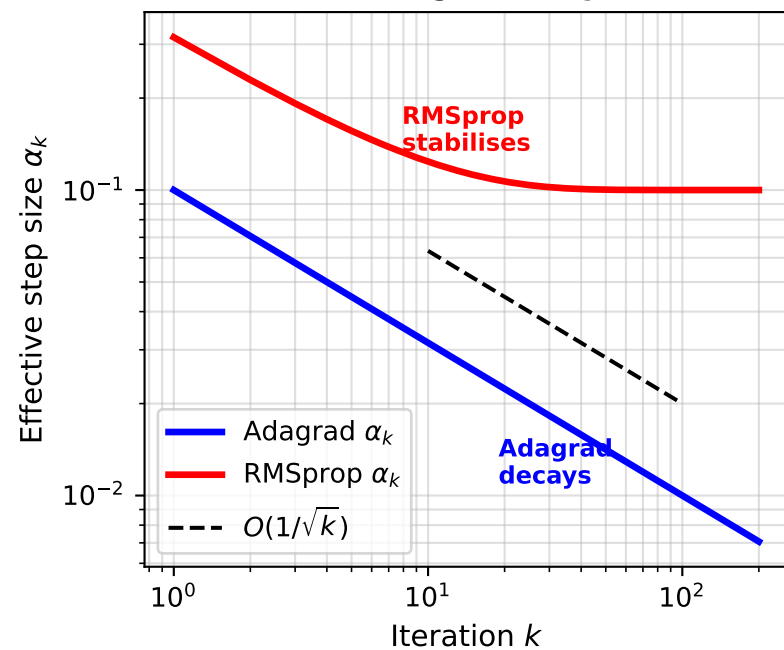
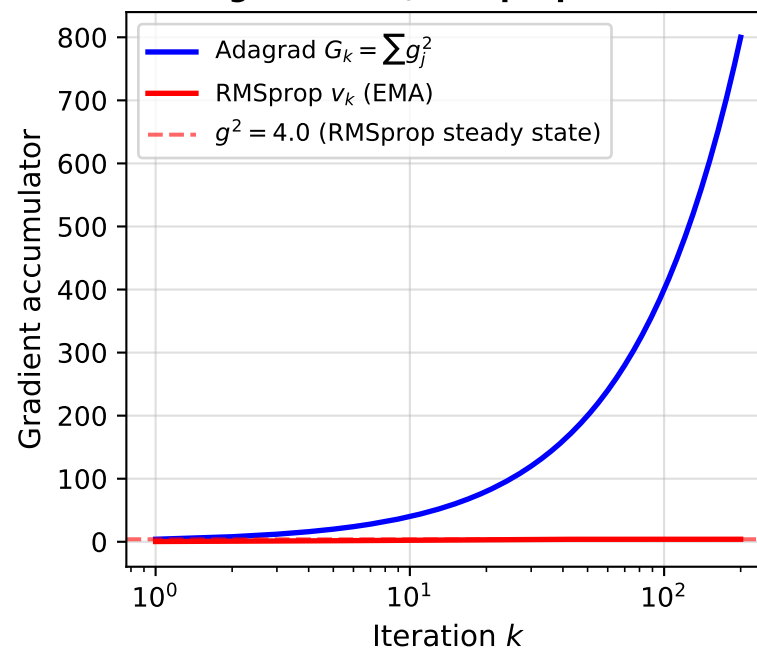


Adagrad vs RMSprop: Step-Size Theory and Coordinate-Wise Adaptation Analysis

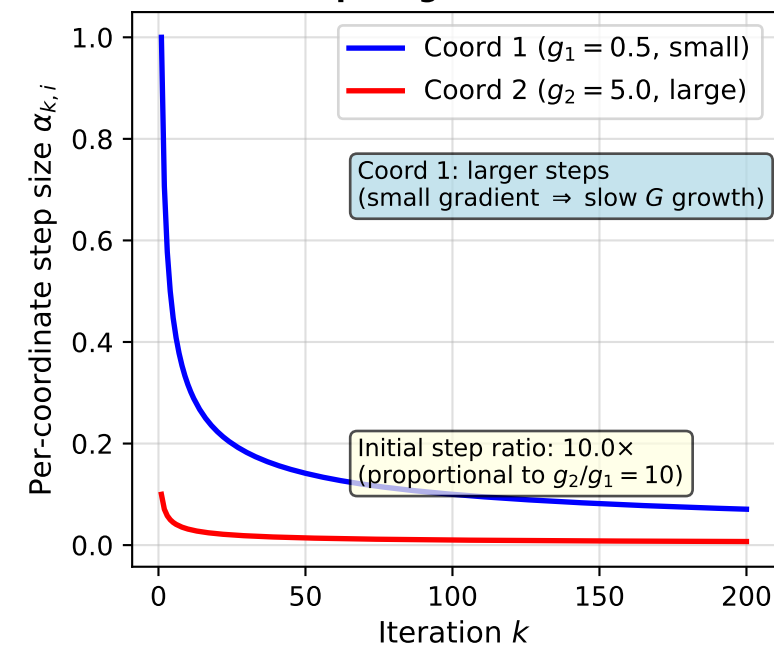
(a) Theoretical step decay
(constant gradient $g = 2$)



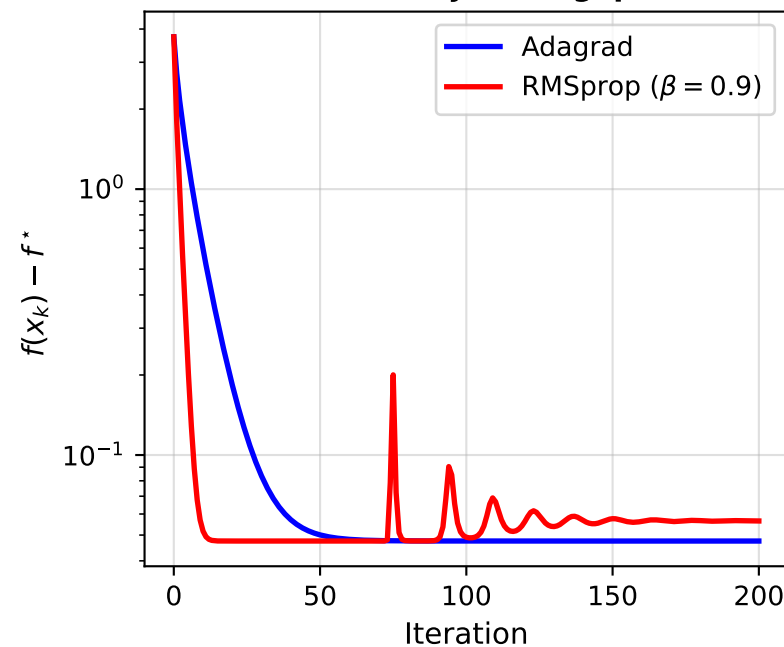
(b) Gradient accumulator growth
(Adagrad linear, RMSprop bounded)



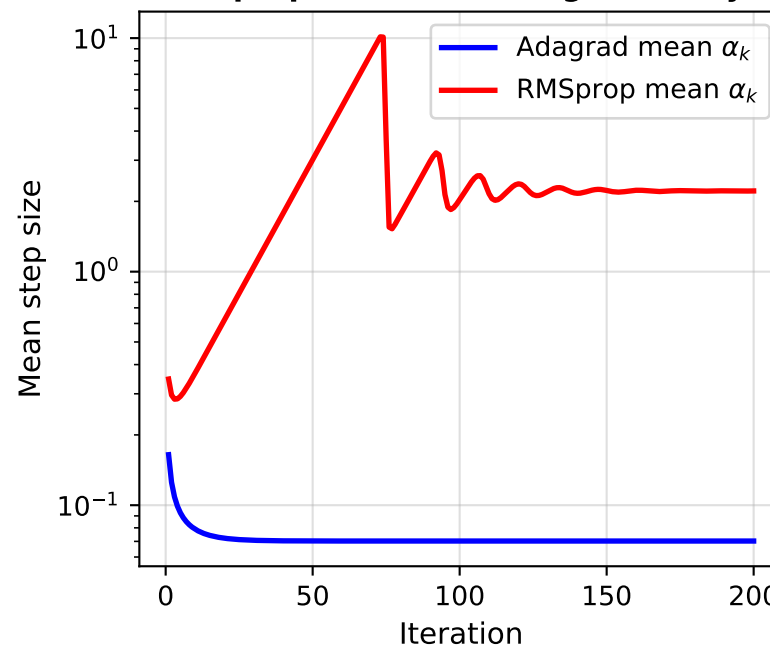
(c) Adagrad coordinate-wise adaptation
(decoupled gradient scales)



(d) Convergence on Benchmark A
(actual objective gap)



(e) Step size evolution on Benchmark A
(RMSprop stabilises, Adagrad decays)



(f) RMSprop memory window
($k = 50$, weight of past gradients)

